LIN7209 – Syntax

Successive cyclic movement

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Long-distance dependencies

- Certain dependencies ($\overline{\text{A}}$-dependencies) can cross clausal boundaries:

(1) Constituent ("*wh*") questions:
Which books does Fatima think [$_{CP}$ Sam said [$_{CP}$ Harisah likes __?]]

(2) Topicalization:
These books, Fatima thinks [$_{CP}$ Sam said [$_{CP}$ Harisah likes __.]]

(3) Relativization:
I grabbed **the books** [$_{CP}$ that Fatima thinks [$_{CP}$ that Sam said [$_{CP}$ Harisah likes __.]]]

- Compare to: raising, passivization, which cannot cross CPs, only non-finite TPs.

Successive cyclicity

- How are long-distance dependencies established?

(4) a. One fell swoop *wh*-movement; no intermediate landing site:
Which books does Fatima think [$_{CP}$ Sam said [$_{CP}$ Harisah likes __?]]

b. Successive cyclic *wh*-movement; intermediate landing sites:
Which books does Fatima think [$_{CP}$ __ Sam said [$_{CP}$ __ Harisah likes __?]]

- Strong evidence the successive-cyclic analysis is superior. But how should we refine it?

Three approaches

- Percolation
- Feature-driven movement
- Feature-less movement

Feature percolation

- No movement. There is never a specific element (*wh*, DP) where the gap is, just a generic gap.
- The information that there is gap is carried up the nodes of the tree ("percolates") until it meets its filler, which *is* a specific element (*wh*-phrase, topicalized DP etc).
- This requires the information to be passed to **each intermediate node**, in a rule-based way.
- Predictions:
  - Any intermediate node may show **traces of the percolated "gap" information** (see it as a feature).
  - However, no intermediate head should show traces that something specific "moved" (no movement!).

Feature-driven movement

- More familiar to us: *something* (*wh*, DP…) starts its life in the gap position, then moves to the top of the tree.
- This requires the element to be **attracted** to the top of the tree; features can be taken to achieve this.
- **Features** are distributed **at regular points along the path of movement**, to make sure the moved element is visible to higher heads (recall phase theory) and makes its way to the top in an "unbounded" way.
- Predictions:
  - Some intermediate node may show **traces of the feature** that's there.
  - There may also be **traces of what is moved** (contrast with the percolation view!).

Feature-less movement

- Again, *something* (*wh*, DP…) starts its life in the gap position, then moves to the top of the tree.
- This time, only one feature is there, near the top of the tree. This solves the issue of intermediate feature placement (in feature-driven approaches), namely, how to make sure features themselves are always regularly distributed?
- Some external constraint forces the moved element to reach that high feature.
- Predictions:
  - No intermediate head should show traces of the high extraction feature (contrast with the feature-driven view!).
  - There may be **traces of what is moved** (contrast with the percolation view!).

Coppe's claims

- The attested reflexes of successive cyclic movement suggest that there are **intermediate traces of the extracted element**, which constitutes evidence for a **syntactic movement approach**, contra the percolation view.
- Additionally, there is evidence for the presence of **extraction features in intermediate spots**, lending support to a **feature-driven intermediate movement approach**, contra the feature-less approach.

Evidence for movement

Partial *wh*-movement

- In some languages, a moving phrase seems to surface in an intermediate position.

(5) Malay (Cole & Hermon 2000):

Ali	memberitahu	kamu	tadi	[$_{\text{CP}}$ **apa**	Fatimah baca __]?
Ali	told	you	just.now	**what**	Fatimah read

'What did Ali tell you just now that Fatimah was reading?'

Partial *wh*-movement continued

- The intermediate *wh* moved in that position and covertly moves higher; evidence from islands.

(6) * Lower relative-clause island:

Ali memberitahu kamu [$_{CP}$ **apa** Mary fikir [$_{CP}$ dia suka [$_{DP}$
Ali told you **what** Mary think he likes
perempuan yang beli __]]]?
woman that buy

'What did Ali tell you that Mary thinks that he likes a woman who bought?'

(7) * Higher relative-clause island

Kamu sayang [$_{DP}$ perempuan yang Ali fikir [$_{CP}$ **apa** telah makan
you love woman that Ali thinks **what** already eat
__]]?

'Who do you love the woman who Ali thinks ate what?'

Multiple realizations

- A copy of a moved *wh*-phrase is realized at the edge of a clause on the path of a long-distance dependency.

(8) German:

Wen	glaubst	du	[$_{\text{CP}}$ **wen**	sie	getroffen	hat]?
who	believe	you	**who**	she	met	has

'Who do you believe she has met?'

- *Wh*-copying is usually limited to constituent questions and relative clauses but sometimes affects all long-distance dependencies, as in Seereer (Baier, 2018) and Dinka (van Urk, 2015).

Stranding

- Parts of the moving phrase (e.g. *all* below) are left in intermediate positions.

(9) West Ulster English (McCloskey, 2000):

a. What **all** did he say [$_{CP}$ he wanted __]?

b. What did he say [$_{CP}$ he wanted **all**]?

c. What did he say [$_{CP}$ **all** he wanted __]?

Interim summary

- **Intermediate copies** can be detected in a variety of ways: exact occurrence (partial *wh*-movement, *wh*-copy), or a part thereof (stranding)
- This is evidence that long-distance dependencies like *wh*-extraction involve **syntactic movement that is successive-cyclic,** i.e. stops at specific intermediate nodes in the structure.
- These observations argue **against a purely percolatory approach**.
- Do they help determine if movement is feature-driven? No!

Evidence for a feature-driven mechanism

Reminder of the predictions

- Under a **feature-driven** approach, intermediate heads attract the moved element because they bear specific features. There should be **morphological evidence for such features**.
- Under a **feature-less** approach, intermediate heads attract the moved element but do not bear specific features. Attraction is forced by an external constraint. **Intermediate attracting heads are not expected to show a special dedicated morphology**.

Irish complementizers

- Irish has a declarative complementizer *go* as well as a complementizer *aL* that surfaces in the context of a long-distance dependency.

(10) a. Creidim [$_{CP}$ **gu-r** inis sé bréag].
believe.1SG C.DCL-PAST tell he lie
'I believe that he told a lie.'

b. an fhilíocht [$_{CP}$ **a** chum sí __].
the poetry C.EXT composed she
'the poetry that she composed.'

- What happens then when the number of clausal boundaries between the filler and the gap increases?

Irish complementizers continued

- In a long-distance dependency, every complementizer on the path of movement must be *aL*!

(11) McCloskey (2002):

an	t-ainm	[$_{\text{CP}}$	**a**	hinnseadh	dúinn	[$_{\text{CP}}$	**a**	bhí	__	ar	an
the	name		C.Ext	was.told	to.us		C.Ext	was		on	the
áit]]											
place											

'the name that we were told was on the place.'

- On top of corroborating the presence of successive-cyclic dependencies, this indicates that the **intermediate C-heads bear specific features**!

Intermediate inversion

- Evidence for extraction features can also be found in the presence of **other morphosyntactic effects** associated with a long-distance dependency.
- For instance, in languages in which movement can be accompanied by inversion, **inversion can happen in intermediate clauses**.

(12) Belfast English (Henry, 1995)

a. Who did John hope [$_{CP}$ **would he** see __]?

b. What did Mary claim [$_{CP}$ **did they** steal __]?

Intermediate agreement

- In a number of languages, the path of long-distance movement is tracked by **agreement with the filler**.
- In Dinka, intermediate movement to the clause edge results in number agreement.

(13)

Yè	kɔ̂ɔc-kó	[$_{CP}$	*Op*	ɛ́-kè-yá		ké	tàak	[$_{CP}$	è	
be	**people**.cs1-which			Ext.Pst-**Pl**-Hab.2Sg		3Pl	think.NF		C	
__	ɛ́-kè-cíi		Áyèn		ké	gâam	gàlàm]]?			
	Ext.Pst-**Pl**-Prf.OV		Ayen.Gen		3Pl	give.NF	pen			

'Which people did (s)he think that Ayen had given a pen to?'

- This makes sense only if **intermediate features trigger agreement** (on top of movement).

Interim summary

- Successive-cyclic movement is accompanied by **morphological markings** (e.g. special complementizers) **and mechanisms** (e.g. inversion, agreement) that are not observed in the absence of movement, and are independent from the specific element that is being moved.
- Thus, a theory based on syntactic movement must incorporate the assumption that **intermediate movements are accompanied by extraction features** as well.

Nature of the intermediate landing sites

Further questions

- We've seen that movement is successive-cyclic, because it leaves intermediate "traces". But **in which exact positions are these traces located?**
- There is consensus that intermediate movement stops at the edge of each clause (Spec-CP). Consistent with the data seen so far.
- Some theories argue that **intermediate movement also stops at the edge of *v*P**. Coppe in particular believes this!
- Let's review evidence for this extra landing site.

West Ulster *all*-stranding may target *v*P

- Some West Ulster English speakers allow *all* to be stranded in Spec-*v*P instead of Spec-CP.

(14) a. What did he [$_{vP}$ **all** do __ on holiday]?

b. What did he [$_{vP}$ **all** say [$_{CP}$ that he did __ on holiday]]?

- Evidence that at least in some language varieties, **the moved element stops in Spec-*v*P (and leaves stuff there)**.

Dinka intermediate agreement targets *v*P

- Remember how *wh*-movement in Dinka leaves a trail of plural markers when the extracted element is plural?
- This also applies to topicalization, and leaves such a marker in Spec-*v*P!

(15)

Ké(ek)	áa-cïi	Áyèn	[[$_{vP}$	ké	tïiŋ].
3Pl	3p-Prf.OV	Ayen.Gen		**3Pl**	see.NF

'Them, Ayen has seen.'

Movement triggers tone changes inside *v*P

- In Asante Twi, every verb on the path of a long-distance dependency is overwritten with a high tone.

(16) Asante Twi (Korsah & Murphy 2019):

a. Kofí kaé [$_{CP}$ sɛ Ám!má kitá bayérɛ̀]
Kofi remember that Ama hold yam
'Kofi remembers that Ama is holding a yam.'

b. Déɛ́n na Kofí káé [$_{CP}$ sɛ Ám!má kítá __]
what Foc Kofi remember that Ama hold
'What does Kofi remember that Ama is holding?'

- This tone change would be surprising if the attracting feature were on C (not local enough to access the main verb). It makes sense if the feature is on *v* instead.

Passamaquoddy intermediate verbal agreement

- Long-distance movement in Passamaquoddy may be accompanied by agreeing participial endings on verbs on the path of the dependency.

(17) Passamaquoddy (Bruening, 2006):

Wot nit pahtoliyas [$_{CP}$ Mali elitahasi-c-**il** [$_{CP}$ eli wen
this that priest Mary IC.think-3Conj-Part.Obv C someone
kisi-komutonom-ac-**il**]].
Perf-rob.AO-3Conj-Part.Obv
'This is the priest that Mary thinks someone robbed.'

- Again this would be surprising if the attracting feature were on C (not local enough to access the main verb). It makes sense if the feature is on *v* instead.

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